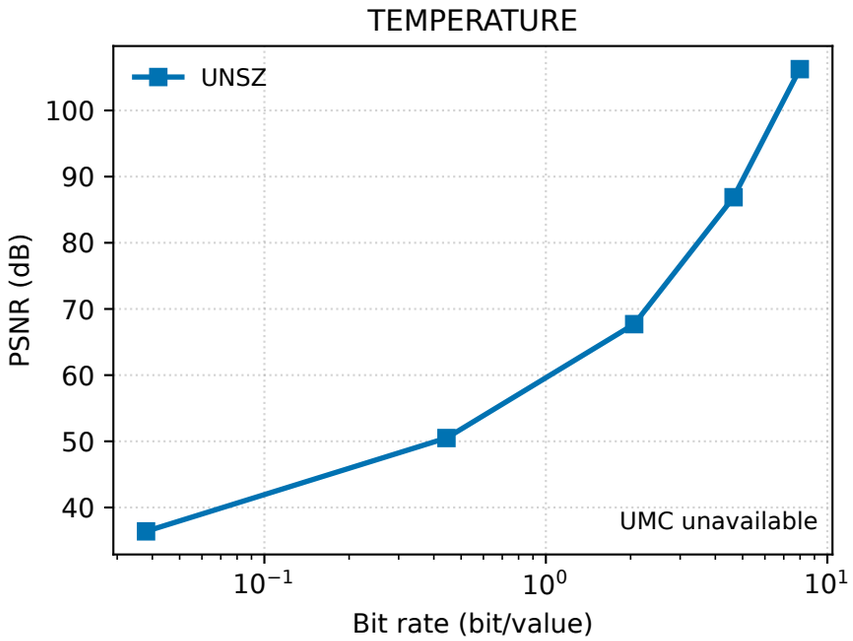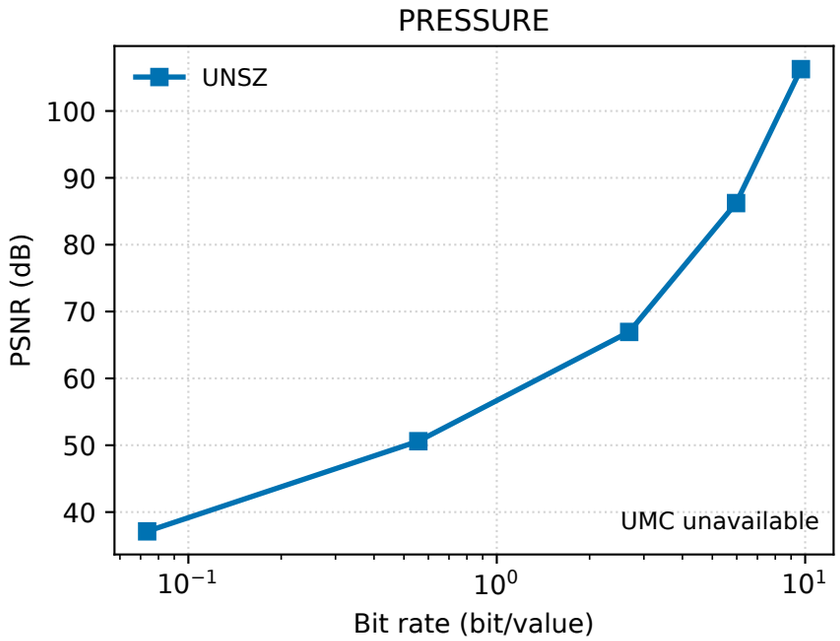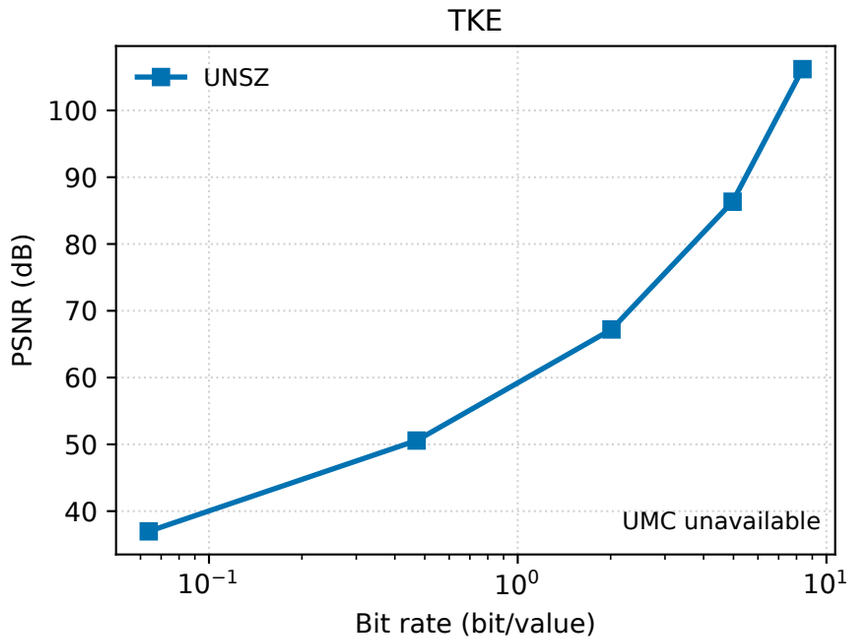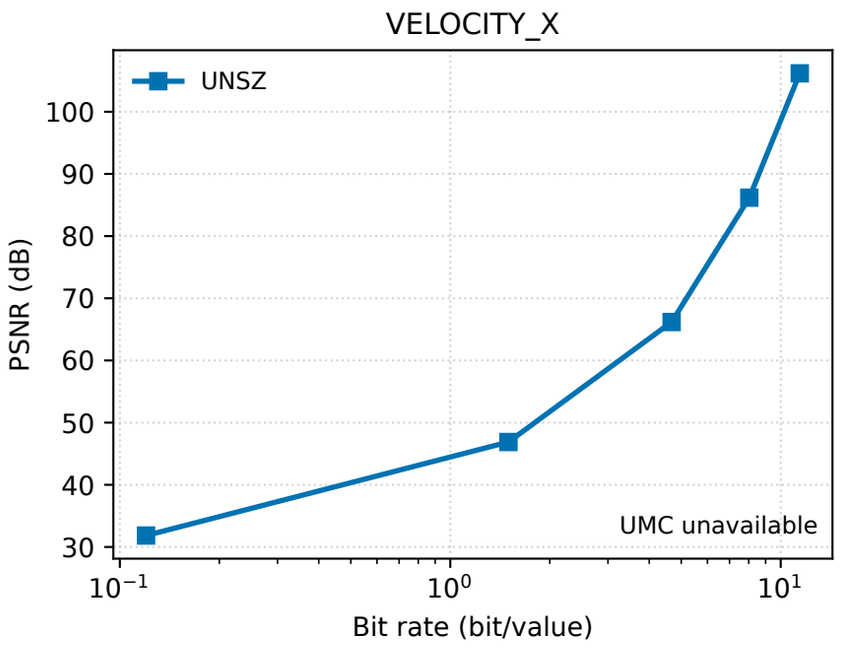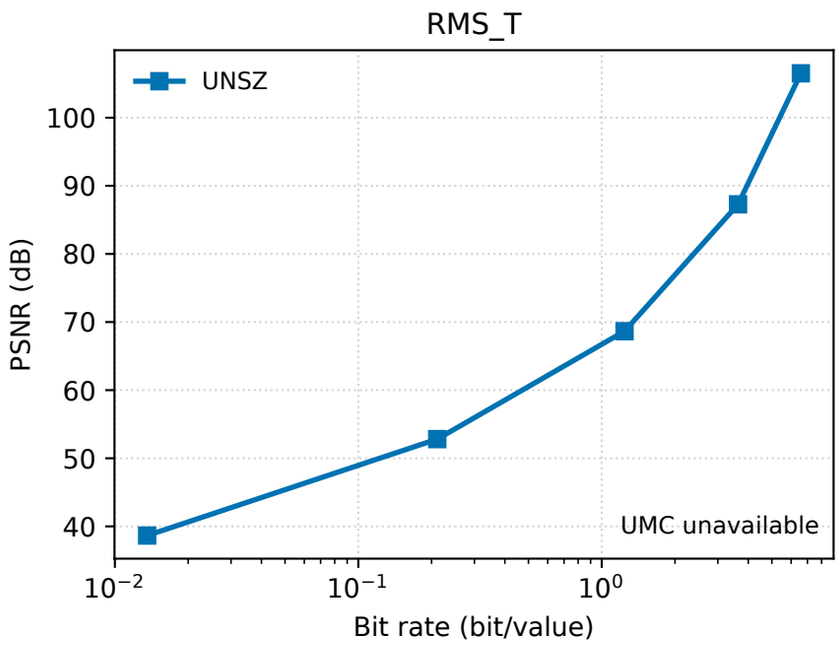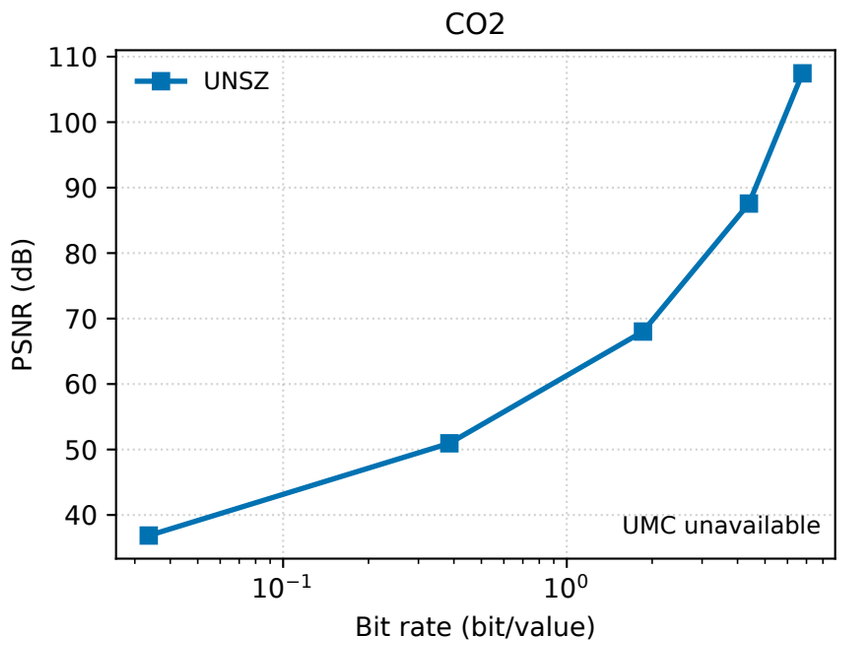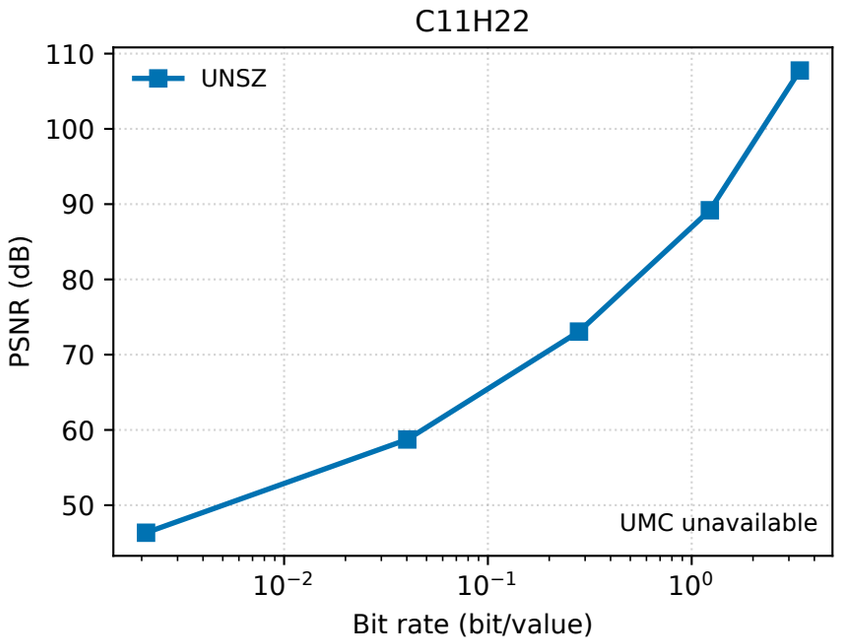taps: rate-distortion